

NIRANJAN BR

ELECTRICAL AND ELECTRONICS ENGINEERING GRADUATE

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PROFESSIONAL SUMMARY

Electrical and Electronics Engineering graduate with hands-on experience in electrical systems, hardware prototyping, and system integration, supported by strong debugging skills. Experienced in building and testing practical projects involving circuit design and embedded systems. Interested in electric vehicle technologies, motor systems, and efficient energy transfer, with a strong willingness to learn and apply engineering concepts in real-world applications.

CORE SKILLS

- Electrical Systems, Circuit Design, Troubleshooting, System Integration, On-grid Solar PV, Embedded Systems
- EV & Power Electronics: Electric Vehicle Technologies, Motor Control, Power Electronics, Battery Management Systems, EV Charging
- Programming: Python, Shell, Bash
- Operating Systems: Linux Server, Windows
- Tools & Platforms: AutoCAD, MATLAB, KiCad, Git, VMware, VS Code
- Professional Skills: Leadership, Team Management, Communication, Coordination
- Languages: English, Malayalam

PROFESSIONAL EXPERIENCE

Solar Technician

Lifestyles, Kollam, Kerala | Aug 2022 – Jul 2023

- Installed and tested 25+ on-grid solar PV systems in residential and commercial setups.
- Achieved 98% first-time fix rate on grid-tied inverter issues and system faults.
- Assisted in system commissioning, wiring, and performance verification.
- Performed routine maintenance and troubleshooting to ensure optimal system performance.

Industrial Training

KSEB, Koorkkenchery, Thrissur, Kerala | Sep 2024

- Completed a 1-week training in substation monitoring, transmission lines, and fault handling. Observed load management and relay operations at the 110kV substation.

Technical Coordinator

IEEE RAS SBC, GEC Idukki | 2024 – 2025

- Organized 6+ events attended by 100+ students. Oversaw logistics and technical setups.

Chairperson

IEEE RAS SBC, GEC Idukki | 2025 – 2026

- Led 15+ technical events, driving a 40% increase in participation.

FOSS Cell Student Ambassador

GEC Idukki | 2025 – 2026

- Promoting open-source culture through workshops, community initiatives, and student engagement activities.

EDUCATION

- B.Tech in Electrical and Electronics Engineering | Government Engineering College, Idukki, Kerala | May 2026
- Diploma in Electrical and Electronics Engineering | Sree Narayana Polytechnic College, Kollam, Kerala | Mar 2022
- THSLC | Technical High School, Ezhukone, Kollam, Kerala | Mar 2019

CERTIFICATIONS

- Python for Data Science | PMMMNMTT – Government of India initiative | Nov 2021

PROJECTS

Dynamic Wireless EV Charging

Developing a 12V dynamic wireless charging prototype using resonant inductive coupling, integrating power converters, control monitoring, and system testing to study efficiency and practical charging performance.

- Implemented transmitter and receiver coils with regulated power transfer.
- Integrated buck and boost converters for 5V–12V voltage conversion.
- Developed real-time voltage monitoring using Arduino.
- Evaluated power transfer efficiency and alignment impact on system performance.

Arduino-Based Fire Fighting Robot

Designed and built an RF-controlled robotic system for fire suppression in hazardous environments.

- Integrated Arduino-based control with motor drivers for navigation and movement.
- Implemented wireless RF communication for remote operation.
- Designed water pump activation system for targeted fire extinguishing.
- Worked on circuit integration, power management, and real-time control.

PhycoGlow

Developed an algal-based lamp system combining illumination, air purification, and aroma diffusion.

- Designed a system utilizing microalgae for natural air purification and oxygen generation.
- Integrated lighting and diffusion mechanisms into a compact functional design.
- Focused on sustainable and eco-friendly product development.
- Patent Application No.: 202641008510 — Filed on 28 January 2026

ScriptSculptor

Toolkit of Bash and Python scripts for automating development workflows and system setup.

- Automated ROM build processes including environment setup, compilation, and cleanup.
- Integrated logging and error handling for improved build reliability.
- Developed scripts for system configuration, dependency management, and workflow optimization.
- Reduced manual effort and improved efficiency in repetitive development tasks.

Voltage OS

Open-source Android custom ROM focused on performance optimization and system stability.

- Actively developing and maintaining since 2021, with continuous updates and improvements.
- Maintained device trees and vendor configurations for Qualcomm-based devices.
- Performed kernel-level debugging and optimizations to improve system performance.
- Built and released 40+ OTA updates with 1000+ downloads.

Meraki

Custom Linux kernel for Android devices with performance and stability enhancements.

- Modified CPU scheduling parameters and thermal profiles for improved efficiency.
- Tuned kernel configurations to balance performance and battery usage.
- Debugged kernel issues and ensured stable builds across device variants.